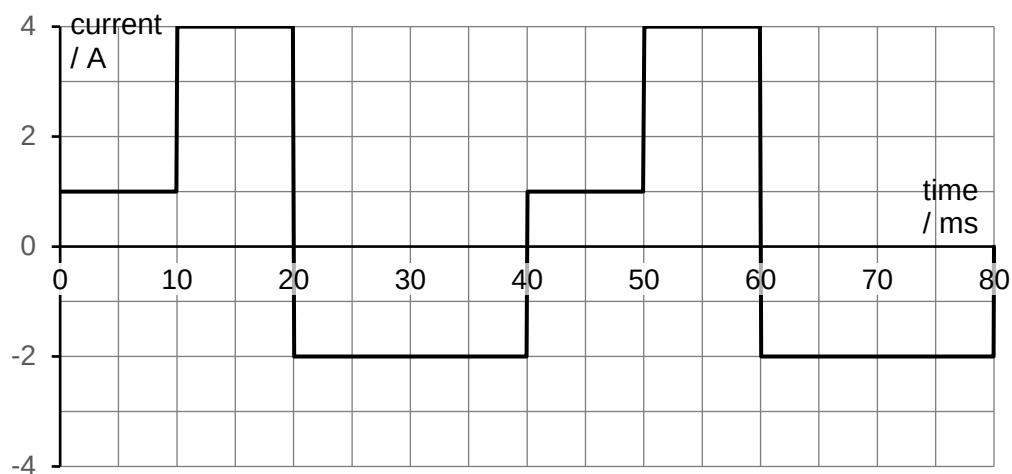


26 The graph below shows how the current in a coil varies with time.



What is the value of a steady current that will dissipate heat in the coil at the same average rate as the current above?

A 1.5 A

B 2.3 A

C 2.5 A

D 2.6 A

Ans: C

$$\text{Area under current}^2 \text{ graph} = (2 \times 1^2 + 2 \times 4^2 + 4 \times 2^2) \times 5 = 250$$

$$\text{Mean current}^2 = 250 / 40 = 6.25 \text{ A}^2$$

$$\text{RMS current} = \sqrt{6.25} = 2.5 \text{ A}$$

27 An ideal transformer steps down the 240 V sinusoidal voltage from the mains to 12 V, which is